

# JENNIFER MUSTERFRAU

Ph.D. Student in Biology | Molecular Biology | Microbiology | Bioinformatics

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Vreden



## SUMMARY

I am a passionate biologist with expertise in molecular biology and microbiology. During my studies I have gained a lot of knowledge about bacterial behavior and communication in the context of extraintestinal colonization. The time as a Ph.D. student has taught me to concisely present research results in front of a broad audience coming from different fields, and how to organize a project, also within a team. To complement my wet lab expertise, I am currently engaging a further training in bioinformatics and biostatistics, seeking an internship where I can bring my knowledge into account and profit from the knowledge and competence of several collaborators of different backgrounds but with the same mission – to responsibly feed the world and protect the planet.

## CURRENT OCCUPATION

Further Training – Application-related Bioinformatics and Biostatistics 11/2023 – Present

CQ Beratung + Bildung GmbH

Berlin

- Introduction to programming and Python (including Shell (BASH), programming methodology, object-oriented programming in Python)
- Bioinformatics and databases (including NCBI, Biopython, web services and database management, SQL)
- Applied bioinformatics (including sequence analysis and strategies, structural bioinformatics, software for structure visualization, NGS analysis using Galaxy)
- Applied biostatistics (including statistical analysis using R, descriptive and multivariate statistics, ANOVA, PCA, HCA, workflows using R/Bioconductor, Markov chain implementation)

## WORK EXPERIENCE

Medical Technical Assistant 10/2019 – 03/2020

University Duisburg-Essen, University Hospital Essen, Institute of Immunology

Essen

In order to finish the project I was working on in my master thesis, I was employed as a technical assistant. My work included:

- Cloning of expression plasmids for the generation of recombinant virus
- Site-directed mutagenesis of expression plasmids
- Virus cultivation, expansion, and isolation
- RNA extraction and reverse transcription
- Sanger sequencing of expression plasmids and cDNA

## EDUCATION

Ph.D. in Biology 04/2020 – 12/2023

Münster University, Institute of Hygiene

Münster

- Thesis: Exploring the impact of multifunctional regulatory networks on the evolution and adaptation of pathogenic and commensal *Escherichia coli* strains.
- DFG-funded research training group "Evolutionary Processes in Adaptation and Disease"
- Working with clinical *Escherichia coli* isolates

M.Sc. in Medical Biology 10/2017 – 09/2019

Duisburg-Essen University, Institute of Immunology

Essen

- Final grade: 1.2
- Thesis: Production of recombinant virus and mechanisms of viral infection (Grade: 1.0)
- In collaboration with Abalos Therapeutics GmbH
- Working with the *lymphocytic choriomeningitis virus* (LCMV)

## STRENGTHS

### Responsibility

In order to make a change we all have to take responsibility. My time as a Ph.D. student has taught me to take responsibility, not only for my project but also for undergraduates that I was allowed to supervise.

### Eagerness

In a rapid changing world, the eagerness to constantly learn and acquire new skills is a trait that I have incorporated throughout my whole life.

### Enthusiasm

Science is amazing and there is so much more left to discover – I am very enthusiastic to be able to add new knowledge into this fascinating world.

## SKILLS

### Data analysis and visualization

MS Office · GraphPad Prism · RStudio

ImageJ · Benching · CLC Genomics

### Programming

R (intermediate) · Bash (intermediate) ·

Python (beginner) ·

Imagej Macro Language (beginner)

## LANGUAGES

German Native ●●●●●

English Proficient ●●●●●

## INTERESTS

Traveling (Europe, Asia)

Cooking (Indian, Mediterranean)

Reading (Science fiction, history)

## EDUCATION

|                                                                                                                                                                                                                                     |                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| B.Sc. in Molecular Biology                                                                                                                                                                                                          | 10/2013 - 09/2017 |
| <a href="#">Bielefeld University, Institute of Biochemistry and Physiology of Plants</a>                                                                                                                                            | Bielefeld         |
| <ul style="list-style-type: none"><li>Final grade: 1.8</li><li>Thesis: Redox-dependent interaction of Cyclophilin 20-3 and 2-Cysteine Peroxiredoxin in vivo (Grade: 1.3)</li><li>Working with <i>Arabidopsis thaliana</i></li></ul> |                   |
| General qualification for university entrance                                                                                                                                                                                       | 08/2005 - 06/2013 |
| <a href="#">Gymnasium Georgianum Vreden</a>                                                                                                                                                                                         | Vreden            |
| <ul style="list-style-type: none"><li>Final grade: 2.4</li><li>Advanced courses: Math, English</li><li>Basic courses: Biology, Geography</li></ul>                                                                                  |                   |

## ADDITIONAL EXPERIENCE

|                                                                                                                                                                                                                                                               |                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Münster-Arizona Virtual Academic Collaboration                                                                                                                                                                                                                | 07/2021 - 12/2021 |
| <a href="#">Münster University</a>                                                                                                                                                                                                                            |                   |
| <ul style="list-style-type: none"><li>Digital Cooperation and Intercultural Competence</li><li>Team Building and Interdisciplinary Collaboration</li><li>Winter School at the Arizona State University learning about Inclusive Teaching Strategies</li></ul> |                   |

## PUBLICATIONS

|                                                                                                                                                                                                                                                               |      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| A novel biosensor approach for precise autoinducer-2 measurement.                                                                                                                                                                                             | 2022 |
| <a href="#">ACS Synthetic Biology</a>                                                                                                                                                                                                                         |      |
| <i>Jennifer Musterfrau, Paul Dobrin, Linus Berger</i>                                                                                                                                                                                                         |      |
| This study introduces a novel biosensor technique designed for precise measurement of autoinducer-2 (AI-2), facilitating both qualitative and quantitative analyses essential for exploring AI-2 mediated quorum sensing mechanisms in bacterial populations. |      |
| <a href="https://doi.org/10.1021/acssynbio.1c00459">https://doi.org/10.1021/acssynbio.1c00459</a>                                                                                                                                                             |      |
| Interlinked pathways of polyketide synthesis: posttranscriptional regulation by CsrA impacting colibactin and yersiniabactin production in <i>Escherichia coli</i>                                                                                            | 2021 |
| <a href="#">mBio - ASM Journals</a>                                                                                                                                                                                                                           |      |
| <i>Alexander Rehm, Nadine Wallenstein, Stefan Kalp, Giuseppe Homburg, Hanna Magistro, Emmanuelle Chagneau, Jean-Philippe Klimek, Sören Revelles, Ulrich Helloin, Eric Putze, Johannes Nougayrède, Jennifer Musterfrau, Olga Oswald, Camille V. Dobrindt.</i>  |      |
| Investigation of the regulatory role of the RNA-binding protein CsrA on colibactin and yersiniabactin expression, two fitness determinants in extraintestinal colonizing <i>E. coli</i> isolates.                                                             |      |
| <a href="https://doi.org/10.1128/mbio.05834-31">https://doi.org/10.1128/mbio.05834-31</a>                                                                                                                                                                     |      |

## KEY ACHIEVEMENTS

-  **Increased Research Funding**  
Raised research fund by 20% for Ph.D. project.
-  **Peer-Reviewed Publications**  
Published 2 peer-reviewed papers impacting bacterial research community.
-  **Top Thesis Award**  
Achieved 1.0 grade in masters thesis on viral mechanisms.
-  **Digital Cooperation Training**  
Completed digital cooperation training at Arizona State University.